What SLEEP Was, What the Evidence Forced, and What It Became

SLEEP: Final Report — the Complete Arc, April to August 2026

Supersedes the 8 August final report; covers the full programme from the original system through the localisation repair and the four-model validation matrix

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Professor Guha,

This report replaces the 8 August final and tells the programme's story whole, in the order the evidence arrived: the system as originally built and the failure it exhibited (§1); the review that demanded rigour (§2); the replication programme that made the failure universal (§3); the extensions we added on our own initiative, including a pre-registered repair attempt that failed informatively (§4); the re-examination that located the failure's three stacked causes (§5); the pre-registered four-model validation of the repaired system (§6); and where SLEEP now stands, with the questions on which we seek your judgement (§7).

The one-line arc: **a free-form recall floor of** 0.006 **that survived every within-paradigm repair rose to** 0.750 **once the consolidation write was moved to where the transformer actually stores facts** — and the repaired system now beats both naive fine-tuning and a matched EWC baseline in its design regime, under gates and predictions committed before the experiments ran. Every number in this report traces to a machine-readable run file in the project repository.

1. What SLEEP was

1.1 The problem and the biological bet

Large language models exhibit a form of architectural anterograde amnesia: weights are frozen at deployment, and anything acquired during inference lives only in the context window. Retrieval augmentation treats this as plumbing — a database lookup appended to every query — rather than as a representational gap. Biology faces the same problem and solves it with a structured pipeline: surprising experiences are tagged at the synapse by prediction error (synaptic tagging and capture); tags compete for a limited pool of plasticity-related proteins; winners are replayed by the hippocampus during sleep, gradually training the neocortex without overwriting it (complementary learning systems). The output specification of that pipeline — continuous, selective, single-exposure learning without catastrophic forgetting — is exactly what an LLM lacks. SLEEP's bet was that implementing the pipeline faithfully would transfer its properties.

1.2 The system as built

SLEEP augments a frozen pretrained transformer with five interacting components:

- **Tagging layer** — flags surprising spans during normal inference via an adaptive z-score on per-token cross-entropy ($\kappa = 1.5$, EMA $\beta = 0.99$), merging adjacent flags into spans and attaching decay/reinforcement dynamics.
- **KV memory bank (W_{fast})** — one-shot episodic storage written directly into attention: pre-RoPE keys and values for the full episode containing a tag, injected at negative position IDs with per-query top-k retrieval gating.
- **PRP allocator** — a budgeted competition over tags scoring surprise, access, cross-reference density, and recency; only allocated tags become consolidation candidates.
- **Sleep engine** — a five-phase cycle (generate replays → quality-check → train → perplexity-check → validate and clean up) that trains a LoRA consolidation adapter (W_{cons}, rank 16) on replayed content, then clears the bank.
- **Safety machinery** — four simultaneous mechanisms: a Fisher-weighted EWC penalty, a hard per-parameter clip $\delta_{\max}$ against the pre-cycle checkpoint, 1:1 interleaving with self-generated "old knowledge," and per-candidate validation with rollback.

Discipline was front-loaded: **36 design questions were resolved in a pre-registered formalization before any code was written**; every hyperparameter default traces to it, and every later deviation carries a dated logbook entry. The implementation grew to ≈4,000 lines across eight modules, with (by programme end) 410 unit and integration tests — including ground-truth equality tests on every critical numerical path, a practice that repeatedly caught silent-failure bugs (a missing layer-norm in K/V extraction; a cache-ordering error duplicating memory each generation step) before they could contaminate results.

1.3 The first campaign and the April headline

The benchmark: 200 synthetic single-sentence facts across 10 templates, with fictional entities so that pretraining cannot supply the answers — if the model knows a fact, it learned it from us, in one exposure. Three metric families: multiple-choice accuracy (MC; recognition against three same-template distractors, chance 0.25), free-form Delayed Recall Accuracy (DRA; keyword recall from a question prompt — the real test), and Base Capability Preservation (BCP; perplexity ratio after/before on held-out controls, 1.0 = unharmed). At evaluation the memory bank is disabled: only what reached the weights counts.

Three April findings framed everything after:

1. **The fast-weight substrate had to change.** LoRA fast weights at the formalization's $\alpha_{\text{fast}} = 10^{-4}$ produce no detectable encoding from single exposures (below the bfloat16 precision floor); direct-write KV injection with top-k gating does: tagged-fact MC 0.28 vs. untagged 0.12 (+0.16) at BCP 1.08. Without gating, an injected bank destroys the model (BCP ≈ 97).
2. **Recognition without recall.** The same memories that bias recognition by +0.16 yield cloze accuracy 0.00 and free-form recall 0.00. Attention-injected memory acts as a small logit perturbation: enough to tilt a four-way comparison, overwhelmed across fifty autoregressive steps against a prior trained on trillions of tokens.
3. **Consolidation stalled even on perfect data.** Replacing generative replay with the original fact tokens (eliminating replay quality as a confound) left recall at DRA $= 0.006 \pm 0.003$ under default safety; a four-point relaxation sweep traced a monotonic frontier with no operating point

achieving DRA > 0.05 at BCP < 1.05. Meanwhile the engine's internal surprise-based validator kept certifying ∼39 consolidations per run.

A fourth correction from this period is worth recording because it validated the proposal's own framing: **tags are pointers, not storage.** The formalization stored only the K/V of tagged tokens; tagged tokens are short novel-entity mentions that capture the *location* of a surprise without its *content*, and replays seeded from them fabricated plausible-sounding values ("$32.6 million") not present in the source. Expanding storage to the full episode containing the tag raised the internal consolidation rate from 40% to 62% and improved BCP from 1.06 to 1.008 — the implementation must store what the pointer indexes.

Alongside these: an early three-cycle comparison suggested SLEEP preserved capability ≈ 2× better than naive fine-tuning — a claim that would later fail replication and be withdrawn (§3.3). At this stage, everything rested on one model (Qwen2.5-7B) and one seed.

2. What the review named

Your review of that first evidence, together with the external (IBM) review, converged on eleven items in three priorities — in one sentence: *the findings are interesting, but nothing is replicated, nothing is compared properly, and the repair the diagnosis implies has not been tried.*

#	Item	The gap it named
1	Multiple random seeds	every number was a single run
2	Extended multi-cycle (10–20)	three cycles cannot show a plateau
3	More models and precisions	everything was Qwen-7B bfloat16
4	Standard benchmarks	synthetic facts are not comparable
5	Missing baselines	RAG named in the introduction, never measured
6	Fix internal validation	the system graded its own homework
7	Tighten biological framing	biology motivated but did not constrain
8	Retrieval-aware fine-tune	the obvious repair was "future work"
9–11	Failure table, calibration plot, repo	packaging and reproducibility

Items 1–3, 5, 6, and 8 restructured the programme (§3–§4); item 4 produced tested loaders for LAMA, CounterFact, and post-cutoff facts whose large-scale runs remain honestly pending; items 7 and 9–11 reshaped the paper.

3. Hostile replication: what we did, what it found

3.1 Infrastructure first

Three pieces of infrastructure were built before any replication number was collected:

- **Independent-process seeding.** A single entry point seeds every RNG (Python, NumPy, Torch CPU/CUDA, hash seed); multi-seed experiments run one seed per process so model initialisation, tagging decisions, replay sampling, and option ordering vary independently; an aggregating harness reports mean ± SD.
- **The ground-truth verification gate.** Before any experiment on a new architecture, three checks run against the real model: an empty bank changes nothing (bit-identical logits), a

populated bank changes output, and disabling restores baseline exactly. The gate **rejected two defective configurations during the programme** — a config-encoding fault and a Qwen-specific attention attribute that would have corrupted every Llama and Mistral run — both fixed before a single invalid number was produced. After repair, Llama's identity check passed at $\max|\text{diff}| = 0$.

- **Two-stage validation** (review item #6). The surprise-reduction proxy was retained as a cheap pre-filter and a second gate added: a deterministic free-form mini-recall test under greedy decoding. A consolidation counts only if both pass, and both numbers are logged.

3.2 The diagnosis becomes universal

The full five-stage protocol was then repeated, five seeds per model, on Qwen2.5-7B, Qwen2.5-1.5B, Llama-3.1-8B, and Mistral-7B:

	Qwen-7B	Qwen-1.5B	Llama-8B	Mistral-7B
Recall from consolidated weights (DRA, 5 seeds)	0.006	0.007	0.006	0.007
Recall with the fact in context	0.760	0.700	0.987	0.927
RAG (top-3 retrieval)	0.213	0.460	0.520	0.563
EWC-only DRA (BCP)	0.070 (1.12)	0.033 (1.43)	0.073 (1.47)	0.157 (2.53)
Proxy confirmations / recall-gate passes	38.8/0.4	23.8/0.0	9.2/0.6	4.0/0.0

Four conclusions, each now cross-family:

1. **The recall floor is universal** — ≈ 0.006 everywhere, from models that follow the same facts in context at up to 0.99. Not a comprehension failure; a property of the mechanism. The weaker-prior hypothesis (a smaller model might let memory steer more easily) is falsified at 1.5B.
2. **Proxy self-validation is near-totally wrong everywhere**: 93–100% of the consolidations the surprise proxy certifies fail the direct recall gate. Families differ greatly in how many candidates they allocate (Mistral a tenth of Qwen); the false-confirmation structure is invariant.
3. **RAG's shortfall against in-context is retrieval failure**, not comprehension — relevant to the paper's framing of RAG as bound rather than baseline.
4. **EWC alone embarrassed the pipeline**: 0.070 recall at BCP 1.12 on Qwen-7B — a better recall-per-damage ratio than any full SLEEP configuration. A textbook regularizer, one line of mathematics, beating a five-component architecture. This number shaped everything that followed.

A float32 control isolated the precision question: the original $\alpha_{\text{slow}} = 10^{-5}$ fails in *any* precision (loss moves $\sim$1% in float32, zero validation passes), so the formalization default had two separable defects — a bfloat16 representability floor *and* an under-scaled magnitude at 7B.

3.3 The ten-cycle programme and the withdrawal of the preservation claim

The interim position (single seed, ten cycles) was that naive fine-tuning overtakes SLEEP by cycle ten. Nine seeded run-pairs across the four models showed that position was itself under-sampled:

Model	Seed	SLEEP BCP@10	Naive BCP@10	SLEEP DRA	Naive DRA
Qwen-7B	0	4.53	3.36	0.050	0.085
Qwen-7B	1	10.20	6.42	0.032	0.080
Qwen-7B	2	2.71	5.44	0.057	0.078
Qwen-1.5B	0	15.96	21.01	0.045	0.035
Qwen-1.5B	1	230.1	45.83	0.012	0.048
Llama-8B	0	4.83	6.09	0.032	0.083
Llama-8B	1	1530.9	7.11	0.045	0.075
Mistral-7B	0	3.22	2.90	0.008	0.192
Mistral-7B	1	1.64	3.72	0.003	0.160

Three regularities held in all nine pairs: SLEEP's damage always follows a step-then-plateau trajectory; naive's recall is higher in eight of nine; and no run of either arm holds BCP < 1.05 past cycle two. What did *not* replicate is the plateau *level*: $2.7 \to 10.2$ across seeds of the identical Qwen-7B configuration, 4.8 vs. 1,531 on Llama — three orders of magnitude on unchanged hyperparameters, flipping the cycle-ten "winner" seed to seed (SLEEP 4/9, naive 5/9). We therefore **withdrew the three-cycle preservation claim** and replaced it with the stronger methodological finding: *fixed-horizon, single-run comparisons of continual-learning architectures are unreliable at the horizons and seed counts this literature typically reports.* (§6 sharpens this: the lottery turns out to be a property of the *choked and unconstrained* configurations, not of consolidation per se.)

4. What we added beyond the checklist

4.1 The retrieval-aware warm-up: a pre-registered repair that failed informatively

Review item #8 asked for retrieval-aware fine-tuning as a baseline. We promoted it instead to the paper's designed extension: if the recognition–recall gap is a missing *skill* — the frozen model never learned to condition generation on retrieved K/V — then a lightweight warm-up should teach it. Mechanism: split neutral passages (WikiText, disjoint from all evaluation facts) into prefix and continuation; write the prefix into the memory bank; train on the continuation with memory injected — lowering the loss requires routing memory into generation. Trainable component: a per-layer, per-KV-head multiplicative gate on memory values (36 parameters, identity-initialised — installing it is a provable no-op, and the equality test asserting this caught two mis-placements during development), optionally joined by the rank-16 adapter. **The success bar was pre-registered before any run**: DRA 0.10–0.15 at BCP < 1.5.

The result, across sixteen runs (two variants $\times$ two seeds $\times$ four models): **recall never moved** — $|\Delta\text{DRA}| \leq 0.008$ everywhere.

Model	Variant	ΔDRA (2 seeds)	BCP after (2 seeds)
Qwen-7B	gate only	+0.002, +0.002	∼1.00, ∼1.00
Qwen-7B	+ LoRA	+0.003, +0.000	1.73, 3.74
Qwen-1.5B	gate only	+0.002, −0.002	1.13, 1.13
Qwen-1.5B	+ LoRA	+0.002, −0.002	5.83, 19.9
Llama-8B	gate only	−0.007, +0.002	1.78, 1.79
Llama-8B	+ LoRA	−0.008, −0.002	$1.5{\times}10^5$, $1.5{\times}10^6$
Mistral-7B	gate only	+0.008, −0.002	2.77, 3.03
Mistral-7B	+ LoRA	−0.002, −0.005	$1.8{\times}10^4$, $1.0{\times}10^5$

The two variants fail differently and informatively. The gate alone converges to the identity: 36 parameters cannot manufacture the skill. The gate-plus-adapter variant demonstrably *learns the objective* (warm-up loss falls $\sim$1 nat everywhere) and lifts recognition where the model survives (MC $0.220 \rightarrow 0.270$ on Qwen-7B) — yet free-form recall stays at floor, reproducing the dissociation *under intervention* rather than merely observing it. Preservation cost is family-dependent and, at the single untuned learning rate, catastrophic on Llama and Mistral (BCP 10^4–10^6). One side-finding: on Mistral the gate learned to *suppress* memory interference it could not exploit, cutting the cost of carrying a populated bank from $8.9\times$ to $2.8\times$.

4.2 The plateau lottery as a contribution

The lottery of §3.3 deserves its own line because of how it emerged: the review asked for longer horizons and more seeds *separately*; running them *together* revealed what neither would alone. It generalises past SLEEP — any continual-learning comparison reported at one horizon with one seed may be reporting the optimisation-path lottery, not the architecture — and it is, to our knowledge, a new observation about clipped adaptation dynamics.

4.3 Where this left the paper

An honest, universal, well-controlled *diagnosis*: recognition without recall; proxy validation near-totally wrong; the cheap repair a replicated null; long-horizon comparisons seed-unstable; EWC-only embarrassing the compound machinery. The title said "Empirical Limits." The question we could not close: **is consolidation into frozen LLMs impossible — or were we doing it wrong?**

5. The re-examination: three stacked causes

5.1 The pre-mortem

Before accepting the negative result, we ran a pre-mortem on the paper: assume the conclusion is wrong — where would the error be? The exercise re-derived every design choice from first principles against the adjacent literature, with fresh eyes, treating our own defaults as suspect. It converged on one assumption that had never been questioned and two that had been questioned too gently:

1. **Placement was chosen by anatomy, not by evidence.** W_{cons} adapted attention V/O projections in the top third of layers because that mapping "played neocortex" in the biological analogy. The knowledge-editing literature — Geva et al.'s feed-forward key-value memories, ROME's causal tracing, MEMIT's mass edits — locates factual associations in *mid-stack MLP* modules. Every consolidation gradient we had ever applied went to the wrong substrate.

2. **Single wordings teach strings, not facts.** Our own data contained the signature: cloze completion (reproduce the sentence) succeeded while question-answering failed. Memorization without extraction. The correction: train each fact on ≥ 20 deterministic paraphrases spanning declaratives, clefts, appositives, and question–answer forms — generated by seeded slot-filling so the 200 base facts remain bit-identical.

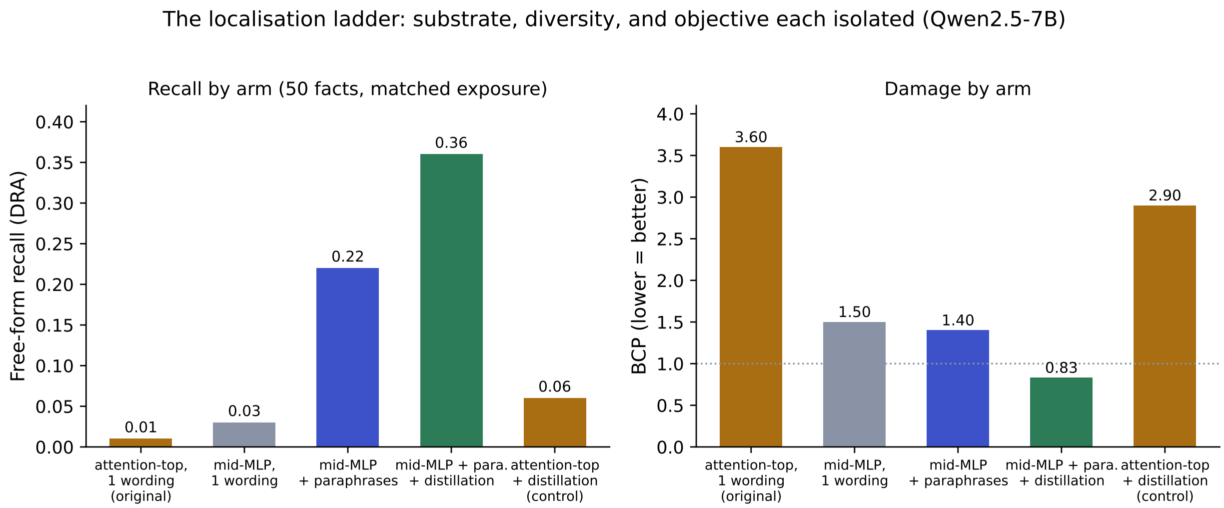
3. **The teacher was already in the building.** Every model answers at 0.70–0.99 with the fact in context. Context distillation makes that the training signal: the teacher is the same model with the fact in its prompt (adapter disabled), and W_{cons} learns to match its token distribution without the context ($T = 2$, KL + $0.5\,$CE, per-token over flattened positions).

Two hygiene corrections rode along: adaptation and injection were *decoupled* in configuration (KV injection keeps its independently verified top-third-attention placement; consolidation moves to mid-stack MLPs), and all knowledge-presence evaluation moved to *greedy decoding* — if the model answers, it knows; no lucky samples.

5.2 The ladder: each factor isolated

Rather than trusting the reasoning, a five-arm ladder isolated each factor on 50 facts at matched exposure (1,200 steps, clipping disabled, greedy evaluation, two models, two seeds):

Arm (Qwen2.5-7B, seed means)	Free-form DRA	Cloze	BCP
attention-top, single wording (original)	0.01	0.44	3.6
mid-MLP, single wording	0.03	0.56	1.2–1.8
mid-MLP + paraphrases	0.22	0.38	1.4
mid-MLP + paraphrases + distillation	**0.36**	0.31	**0.83**
attention-top + distillation (control)	0.06	0.35	2.9



The ladder as a picture: recall by arm (left) and damage by arm (right). The control arm (rightmost) shows the objective cannot rescue the wrong substrate.

Three findings, consistent across both models and seeds. **Placement alone halves background damage in 8 of 8 comparisons** — the same gradient budget, spent in the substrate the editing literature indicates, does less harm and more good. **Paraphrases convert memorization into extraction** — cloze falls (0.56 → 0.38) while recall rises (0.03 → 0.22); the model stops reproducing a token sequence and starts storing a retrievable association. **The in-context teacher is the strongest and gentlest write** — 0.36 at BCP 0.83, sixty times the original arm at better-than-baseline preservation, while the placement control (same objective, old substrate) recovers almost nothing. On Llama the best arm is paraphrase-CE at 0.353. The recall floor was not a law of nature. It was a map error.

5.3 The third cause: the safety machinery erased the write

Wiring the winning recipe into the full autonomous pipeline — tagging, PRP selection, sleep engine, two-stage validation — initially collapsed it back to the floor: trained-subset DRA 0.007/0.007/0.007

across three seeds, zero externally confirmed consolidations. Rather than collecting more identical failures, we killed the queue and A/B-tested the machinery itself. With clip and plasticity scaling relaxed, the identical pipeline reached $\mathbf{0.210 \pm 0.018}$ at BCP 0.90–0.99, with **21–24 of ∼50 selected facts passing the direct greedy recall gate** — the first externally verified autonomous consolidations in the programme. The designed clip ($\delta_{\max} = 0.01$) had been erasing the write *after* training performed it, consistent with the saturation geometry identified in April (A.1). A moderate calibration ($\delta_{\max} = 0.1$, scaling on) keeps both the recall and the protection: 0.189 ± 0.031 at BCP 0.882 ± 0.027 across five seeds — every seed below 1.0, inside the region the diagnosis-era frontier suggested was unreachable.

The original consolidation failure therefore had **three stacked causes** — wrong substrate, wrong surface statistics, over-tight clip — **each alone sufficient to hold recall at the floor**. That is why no single-factor intervention in the diagnosis era ever moved it, and why the failure looked so convincingly structural.

6. The validated system: the four-model matrix

6.1 Pre-registration and conduct

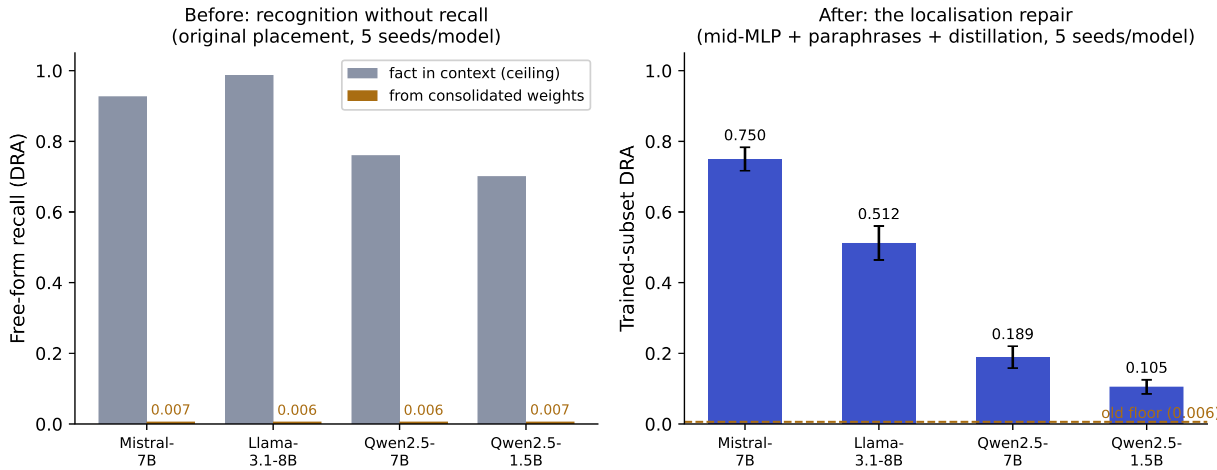
The validation campaign ran 11–12 August on a single A6000 as five staged gates, with the criteria, quantitative predictions, and a stop-loss **committed to the repository before launch**; one extension (a clip ladder for Llama) was registered mid-campaign with its selection rule fixed before its runs executed. Conduct: **35 runs, 19.7 GPU-hours, ≈USD $15.5, zero failures, zero re-runs**; every gate read as written, including the one Llama fails.

D0 (verify). All four mid-MLP configurations passed the three-check ground-truth gate on real weights — including Mistral-7B, whose sliding-window attention exercised the architecture-generalisation fix for the first time (identity max $|\text{diff}| = 0$), and Qwen2.5-1.5B, both first-timers.

D1 (per-family tuning, seed 0; advancement gate: trained-subset DRA ≥ 0.10 at BCP ≤ 1.5). Mistral: distillation 0.750 @ 1.31 **pass**; paraphrase-CE 0.655 @ 2.98 fail — distillation advances (as on Qwen: the in-context-teacher write is also the gentler write). Qwen-1.5B: 0.103 @ 1.03, a thin but genuine pass (10 of 42 selected facts survive the recall gate, against 29 of 31 on Mistral-7B — extraction capacity appears to grow with scale). Llama: the pre-registered clip ladder reduced damage monotonically ($\delta_{\max}$ $0.1 \to 0.05 \to 0.03$: BCP $4.35 \to 3.64 \to 2.14$) with recall degrading only slowly ($0.659 \to 0.593 \to 0.520$), but never reached the bar — **Llama's damage is not clip-dominated**. Per the rule fixed in advance, it runs the matrix at $\delta_{\max} = 0.03$ and is reported as *not yet tuned*, not silently excluded.

6.2 Single cycle, five seeds per model

Model	Recipe	Trained-subset DRA	BCP
Mistral-7B	distil, $\delta_{\max}$=0.1	$\mathbf{0.750 \pm 0.033}$	1.205 ± 0.058
Llama-3.1-8B	par.-CE, $\delta_{\max}$=0.03	0.512 ± 0.048	2.088 ± 0.140
Qwen2.5-7B	distil, $\delta_{\max}$=0.1	0.189 ± 0.031	0.882 ± 0.027
Qwen2.5-1.5B	distil, $\delta_{\max}$=0.1	0.105 ± 0.020	1.043 ± 0.016



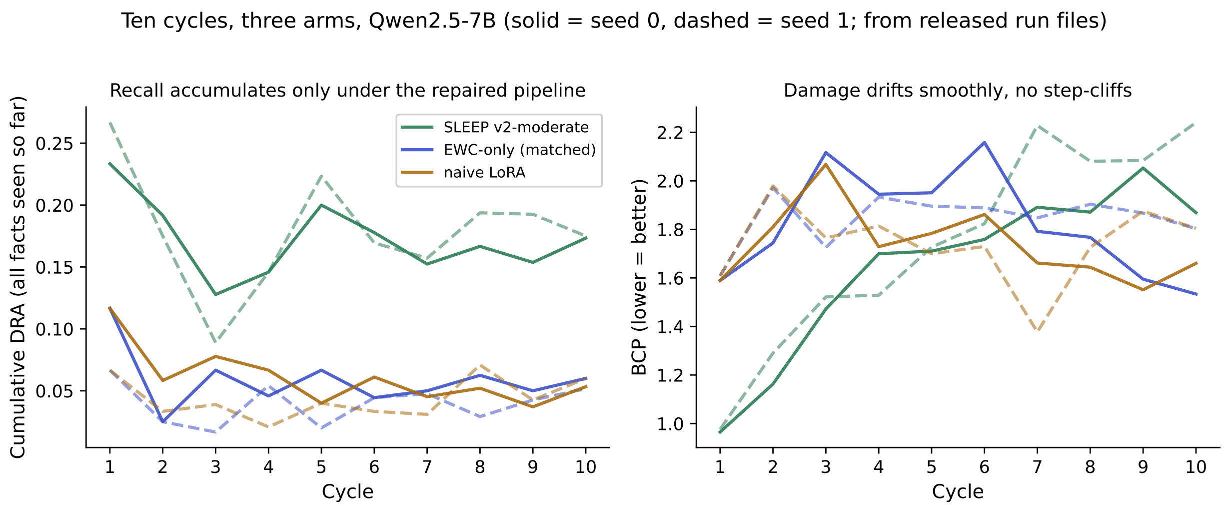
The programme in one figure: the diagnosis-era dissociation (left) against the repaired single-cycle results (right, 5 seeds per model, old floor dashed).

Both pre-registered predictions held: Qwen-7B's mean fell inside the predicted $[0.15, 0.30]$, and every model exceeds the diagnosis-era floor by at least 17× (the weakest by ∼17, the strongest by ∼125). Two observations matter for interpretation. The effect is **strongest off the development model** — Mistral-7B, which the recipe never touched during development, is the best single-cycle result in the programme, arguing against the recipe being overfit to Qwen. And the seed variance is tight everywhere (SD ≤ 0.05): the repaired pipeline is not a lottery.

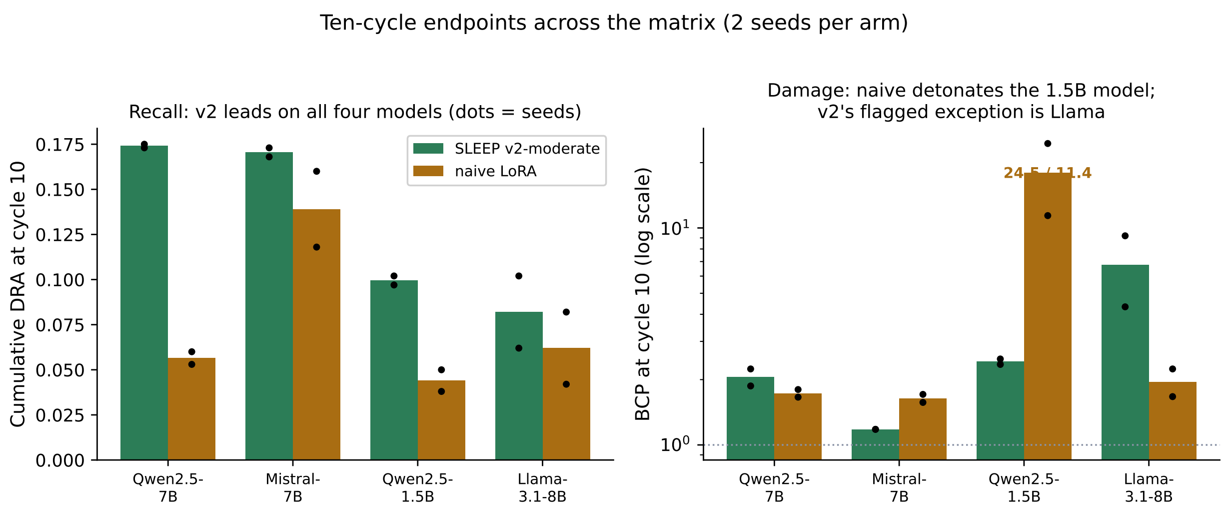
6.3 Ten cycles, both arms, and the EWC showdown

Ten cycles of 20 facts each (400 distillation steps per cycle for the SLEEP arm; the naive arm at the same mid-MLP adapter geometry so placement is never a confound), two seeds per arm; plus a matched EWC-only arm on the two headline families — identical adapter, steps, and learning rate to naive, the *only* difference an online-EWC Fisher penalty ($\lambda = 100$, accumulated per cycle), its accumulation and anchoring verified by ground-truth unit tests before use.

Cycle-10 endpoints (s0/s1)	SLEEP v2-moderate	EWC-only (matched)	Naive LoRA
Qwen2.5-7B	**0.173/0.175** @ 1.87/2.24	0.060/0.052 @ 1.53/1.80	0.053/0.060 @ 1.66/1.80
Mistral-7B	**0.173/0.168** @ 1.18/1.18	—	0.118/0.160 @ 1.71/1.57
Qwen2.5-1.5B	**0.097/0.102** @ 2.35/2.49	—	0.038/0.050 @ **24.5/11.4**
Llama-3.1-8B	0.102/0.062 @ 4.33/9.20	0.075/0.072 @ 2.32/1.79	0.082/0.042 @ 2.24/1.67



Ten-cycle trajectories on Qwen2.5-7B, all three arms, both seeds, plotted from the released run files: the repaired pipeline holds a $\approx 3\times$ separation above both baselines for the entire horizon.



Cycle-10 endpoints across the matrix (dots = individual seeds). Right, log scale: naive fine-tuning detonates the 1.5B model while the constrained arm holds steady; the constrained arm's own flagged exception is Llama.

The pre-registered prediction (v2 > naive on cumulative recall on at least two of the three newly tested models, both seeds) was met on three of three; with Qwen-7B it holds on all four. Findings, each seed-consistent:

1. **The design-intent claim is now real.** The safety-constrained system accumulates more recall than unconstrained fine-tuning over ten cycles on every model — $3\times$ on the anchor model. On Mistral it wins *both axes simultaneously*: more recall and less damage, both seeds.
2. **Right-sized machinery is load-bearing at small scale.** Naive training detonates the 1.5B model — BCP 24.5 on one seed, 11.4 on the other, the plateau lottery reproduced with seed-random blow-up size — while the constrained arm holds ≈ 2.4 on both. The machinery matters most exactly where models are most fragile.
3. **The EWC dissociation.** The penalty gives EWC-only the best damage numbers, but its recall never rises above naive's: *regularisation prevents forgetting; it does not create extraction.*

Preservation and acquisition are different capabilities, and systems scored only on forgetting metrics would rank this experiment's arms incorrectly. SLEEP passes the pre-registered headline gate — 3× both baselines at comparable damage on Qwen-7B.

4. **Stability is SLEEP's quiet signature.** Across-seed spread ≈ 0.005 DRA on every model, against naive's 0.04–0.06; damage drifts smoothly with no step-cliffs. The §3.3 lottery resolves into an asymmetric claim: constrained arms can be made seed-stable; unconstrained arms are the lottery.
5. **Honest exceptions, kept visible.** Llama repeats its single-cycle pattern — more recall than naive on both seeds at damage that compounds unacceptably — and on Llama, EWC-only is competitive with the not-yet-tuned v2 at far better BCP. The Llama column is not a SLEEP win, and the paper says so. One qualitative signature recurs on every model: naive scores higher on *current-cycle* recall while SLEEP retains *earlier* cycles better — memorization versus consolidation, visible in the horizon dimension.

7. Where SLEEP stands, and what we ask

7.1 The claim set

SLEEP set out as a memory system, became a rigorous diagnosis of why the obvious version fails, and is now a working consolidation system with receipts: **a frozen model reads once, sleeps, and answers questions from its own weights** — 0.750 on its best family — **validated across three families and two scales, against real baselines, under pre-registered gates**. The diagnostic findings stand independently: the recognition–recall dissociation for attention-injected memory (unchanged by the repair, which routes knowledge through weights, not injection); the 93–100% proxy-validation failure, which we now treat as a safety-relevant result about self-certifying learning systems; the sixteen-run warm-up null; and the seed-instability of unconstrained continual learning.

The meta-lesson generalises beyond this project: **biological mechanisms transfer — tagging, selective replay, staged consolidation, sleep-phase scheduling — but biological anatomy does not.** The hippocampus-to-top-layers mapping was the single most expensive assumption in the programme; the substrate had to come from the transformer's own causal structure, which the knowledge-editing literature had already mapped. Accordingly, the formalization now carries six empirical amendments — documented as first-class results: the record of what the evidence forced us to unlearn.

Amendment	One-line content
A.1	$\delta_{\max}$ must scale with model size (10^{-3} saturates in ∼30 steps at 7B)
A.2	α_{slow} fails twice over: bfloat16 representability *and* magnitude
A.3	absolute surprise thresholds have a generator–discriminator gap
A.4	surprise-reduction validation has a data-volume floor
A.5	tags are pointers, not storage — store the episode the tag indexes
A.6	**consolidation placement must come from causal structure, not anatomy**

7.2 Named gaps

- **Early-batch fade.** Cumulative recall decays across cycles because the engine rehearses only the current batch. Interleaved rehearsal of prior consolidations — the mechanism biological sleep actually implements — is the clearest next component, and the one we would scope first.

- **The damage leak.** Every arm leaks $\approx +0.12$ BCP per cycle; no configuration holds BCP < 1.05 over ten cycles. Candidate mechanisms: Fisher refresh against the original base model rather than the drifting anchor; consolidation-aware early stopping.
- **Llama.** Recall gains like every other model; damage no tested clip contains. Per-family learning-rate and signal calibration is required, not optional — the matrix provides the harness and the sweep is cheap.
- **Scope.** Synthetic template facts (benchmark loaders built, large-scale runs pending); dense decoders $\leq$8B; two seeds multi-cycle (though every multi-cycle claim held on both, and the constrained arm's variance is demonstrably small).

7.3 The paper and the record

The paper is restructured around the complete arc in two movements — diagnosis (§§1–4 of this report) in full rigour, then repair and validation (§§5–6) — retitled *"Recognition Without Recall: Diagnosing and Repairing Biologically-Inspired Memory Consolidation in Transformers"*: 31 pages, nine figures (the before/after arc now opens the paper), every claim carrying its seeds and its number. Reproducibility is total: the repository holds the implementation (410 tests), the 36-question formalization with its six amendments, the pre-registration documents, and every result file and execution log.

Campaign	When	Scope	Cost
Single-cycle diagnosis	April–May	Qwen-7B: substrate, episodes, Pareto	≈$5
Cross-model replication	early Aug	4 models × 5 stages, float32, baselines	≈$14
Warm-up extension	early Aug	16 runs, 2 variants, 4 models	(incl. above)
Localisation ladder	11 Aug	5 arms × 2 models × 2 seeds	≈$3
Pipeline integration (B/C)	11 Aug	machinery A/B; 10-cycle, 3 arms	≈$5
Validation matrix (D)	11–12 Aug	35 runs, pre-registered, zero failures	≈$15.5
Total		≈95+ seeded runs	≈$43

7.4 Requested input

We would welcome your judgement on three points: **(i)** whether the diagnosis-era first half is correctly weighted against the repair, or should compress further; **(ii)** the target venue and deadline, so the submission draft can be formatted accordingly; and **(iii)** whether the interleaved-rehearsal follow-up — the clearest next mechanism, and a natural continuation of the multi-cycle programme — should be scoped now or after review.

Respectfully submitted — Aditya Tripathi. All runs, gates, and pre-registrations are in the project repository (`logbook/2026-08-11_phase_d_preregistration.md`, `logbook/2026-08-12_phase_d_full_matrix.md`, results under `experiments/results/`). The proposal (7 July), interim report (27 July), and superseded final (8 August) accompany this document.